

Supply Chain SLA Predictability Analysis – Final Report

Roy Arce

1. Problem Statement

Supply chain performance is highly dependent on the ability to meet customer delivery expectations. Delays in shipment can lead to customer dissatisfaction, increased operational costs, and revenue impact.

This project aims to develop a machine learning solution that predicts whether shipments will meet Service Level Agreements (SLA), enabling proactive identification of at-risk orders and improving decision-making across supply chain operations.

2. Model Outcomes or Predictions

This project uses supervised learning (classification) to predict whether a shipment will meet SLA.

The expected output of the model is a binary classification:

- SLA Met
- SLA Missed

The goal is to identify high-risk shipments in advance and support proactive mitigation strategies.

3. Data Acquisition

The dataset was sourced from S/4HANA shipment and order data. It included shipments (orders shipped) and also current backlog (active/open orders)

Key variables include:

- Customer Requested Delivery Date (CRD)

- Actual Shipment Date
- Material Availability Date
- Factory Plant
- Product Line
- Order Priority

The dataset was analyzed to ensure it contains sufficient information to evaluate SLA performance and identify delay drivers.

4. Data Preprocessing / Preparation

a. Data Cleaning

- Removed duplicates and inconsistent records
- Filtered out non-relevant items (e.g., intangible indicator)
- Handled missing values appropriately

b. Train/Test Split

- Data was split into training and testing sets to evaluate model performance

c. Feature Engineering

- Created SLA target variable
- Derived delay and lead-time features
- Encoded categorical variables

5. Modeling

A classification approach was used to predict SLA adherence. Five models including the base model were used and compared to each other for final selection

The final selected model was:

- Logistic Regression

Hyperparameter tuning was performed using:

- Grid Search
- Randomized Search

Cross-validation was used to ensure model stability and prevent overfitting.

6. Model Evaluation

Metrics Used:

- ROC-AUC
- Precision-Recall AUC
- Accuracy

Results:

- ROC-AUC ≈ 0.95
- PR-AUC ≈ 0.997
- Accuracy ≈ 0.89

Due to class imbalance, greater emphasis was placed on ROC-AUC and Precision-Recall metrics.

The final model was selected based on:

- Predictive performance
- Stability across validation

- Interpretability for business use

7. Findings

- Material availability is the strongest driver of delays
- Performance varies across factories
- Operational priorities are not always aligned with outcomes
- Predictive modeling enables early identification of high-risk shipments

8. Conclusion

This project demonstrates how data science can enhance supply chain decision-making by transforming operational data into predictive insights.

The model provides a reliable method to identify SLA risks early, allowing organizations to shift from reactive issue management to proactive intervention.

This approach lays the foundation for scalable solutions such as real-time monitoring, automated alerts, and continuous performance improvement.

9. Next Steps

- Integrate model into dashboards
- Expand dataset with additional historical data
- Enable real-time predictions
- Apply Generative AI for automated reporting